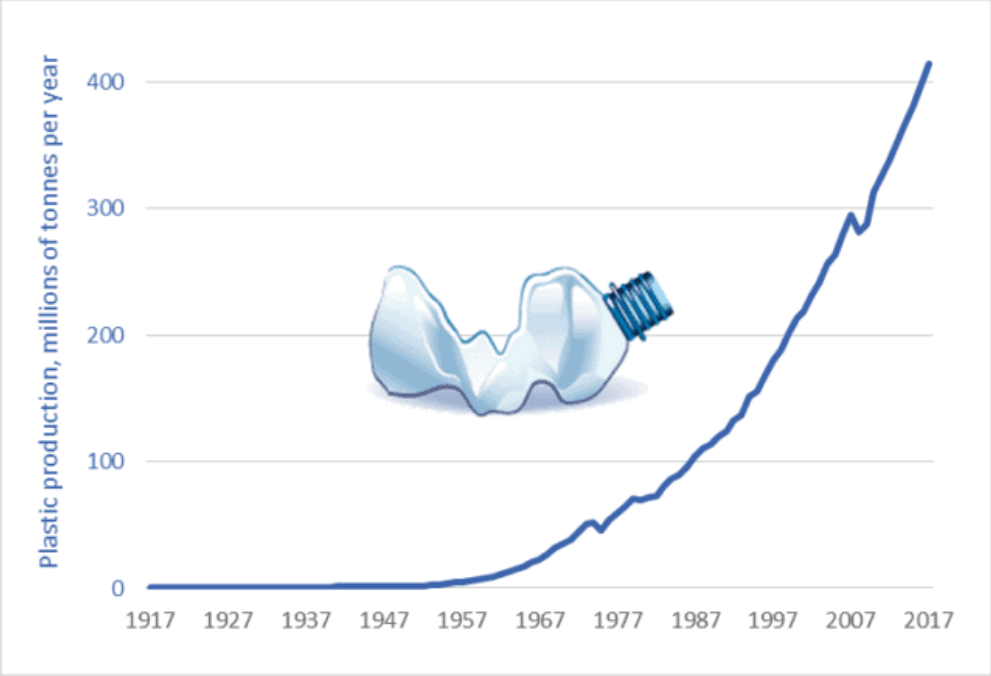
Plastic waste in the Environment

Plastic waste is currently generated at a rate approaching 400 Million Tons year^{-1}

more than a billion kilograms per day!



The vast majority of monomers, such as ethylene and propylene, are derived from fossil hydrocarbons.

Mass production of polymers began only in the 1950s

Plastic production has sharply increased over the last 70 years. In 1950, the world produced just two million tonnes. It now produces over 450 million tonnes.

Plastic has added much value to our lives: it's a cheap, versatile, and sterile material used in various applications, including construction, home appliances, medical instruments, and food packaging.

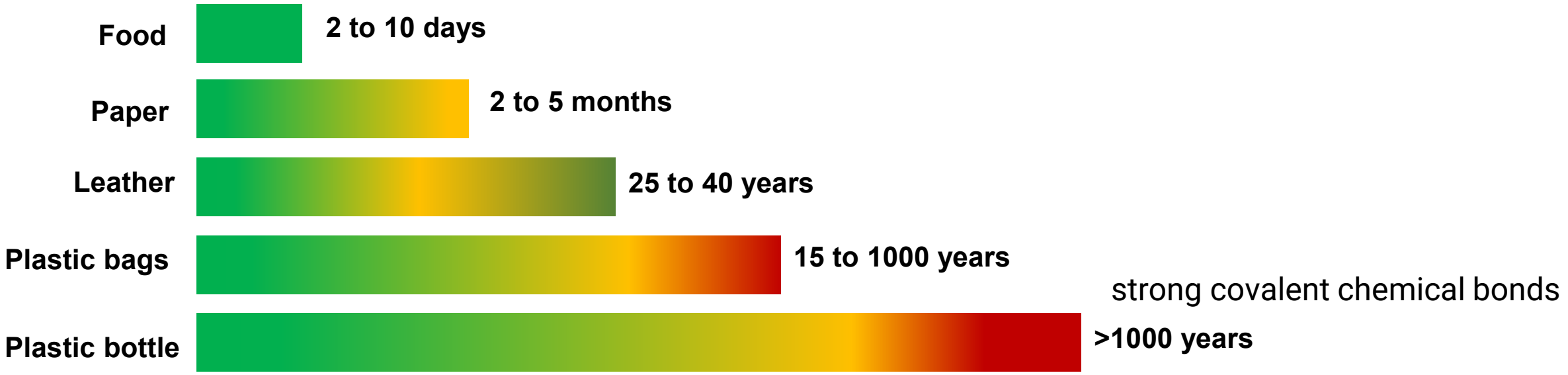
However, when plastic waste is mismanaged — not recycled, incinerated, or kept in sealed landfills — it becomes an environmental pollutant. In particular, one to two million tonnes of plastic enter our oceans yearly, affecting wildlife and ecosystems.

Improving the management of plastic waste across the world — especially in poorer countries, where most of the ocean plastics come from — is therefore critical to tackling this problem.

Organic Chemicals in the Environment

Fate of polymers after use

Comparison of decomposition of plastic waste with other materials



Organic Chemicals in the Environment

Fate of polymers after use

Plastics are forever

Plastics are a product of human ingenuity and innovation—one of civilization's great solutions.
They're lightweight, durable, airtight, decay resistant, inexpensive, and moldable into a huge range of products.

At global level

18% of plastics are recycled,

24% was used for energy, and

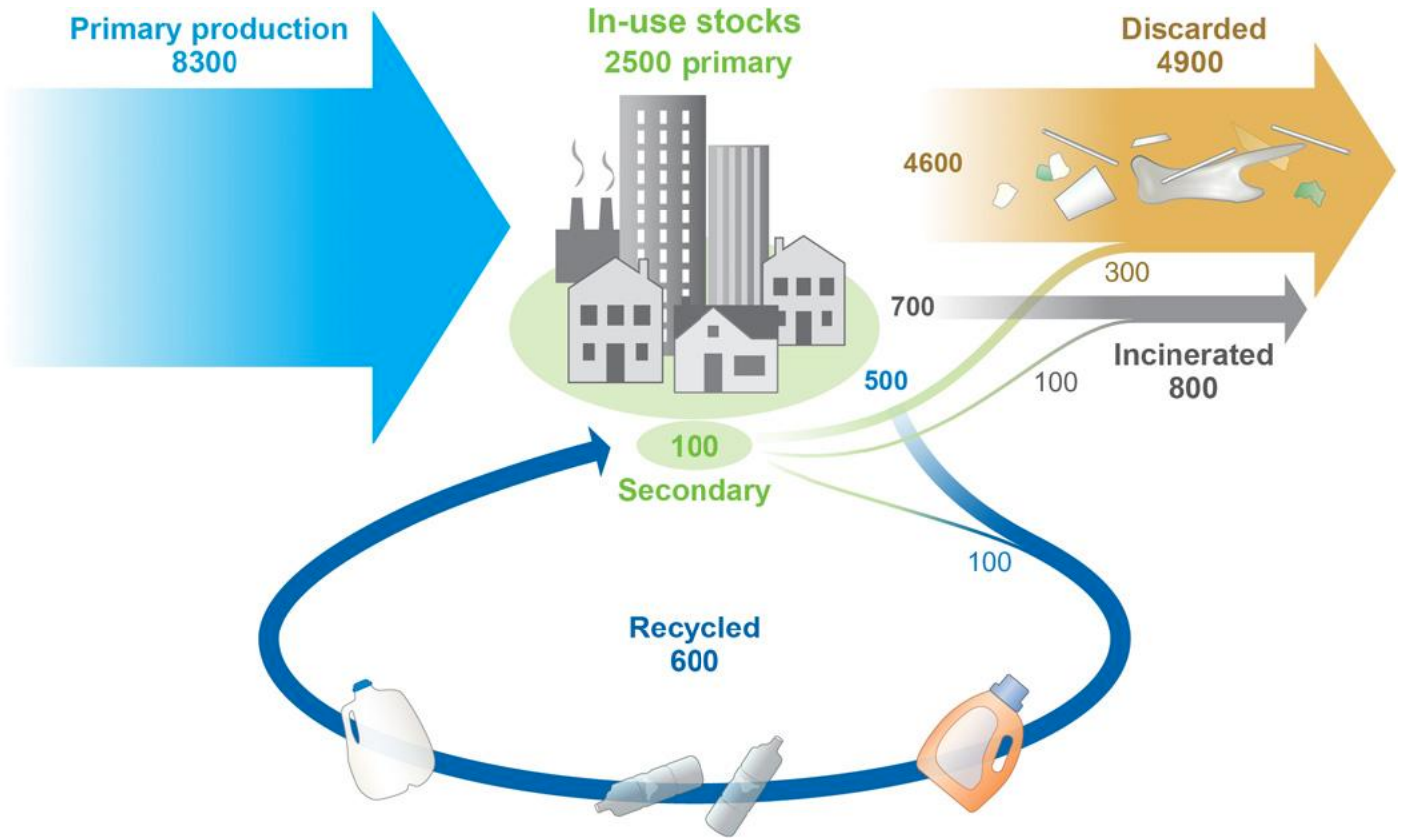
58% was placed in landfills or enter environment

Reference: 10.1126/sciadv.1700782

Organic Chemicals in the Environment

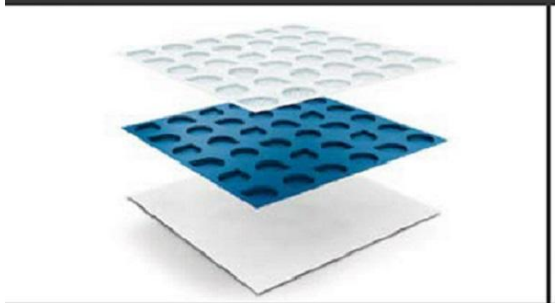
Fate of polymers after use

Global production, use, and fate of polymers (1950 to 2025 in million metric tons)



Reference: 10.1126/sciadv.1700782

Uses Of Plastic In Building Construction


Waterproofing Membranes

Water Tanks

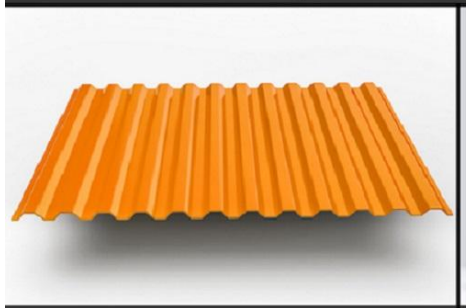
Sealants

Chair & Table

Pipes

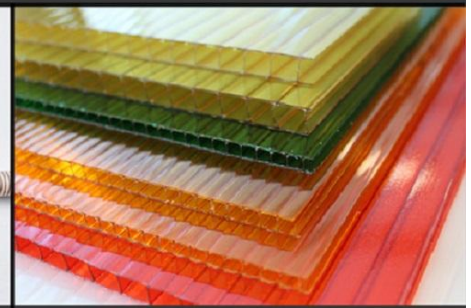
Shutters

False Ceiling Panels


Roofing Sheet

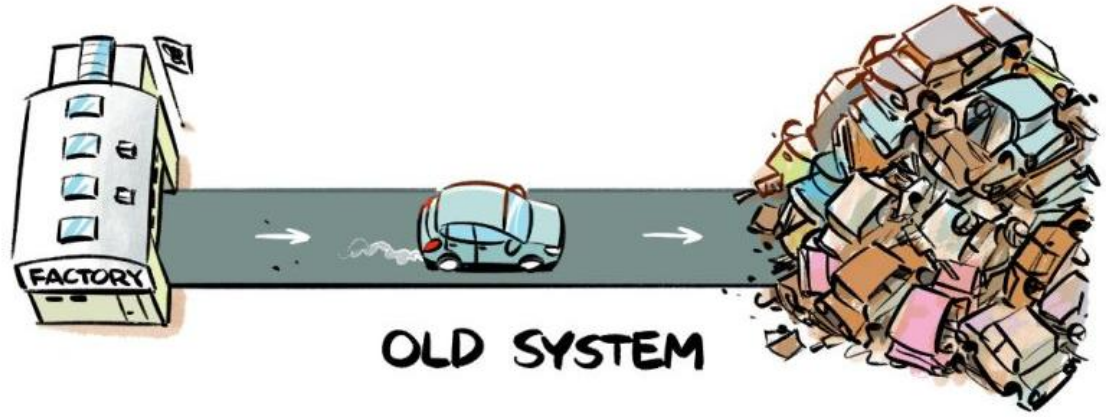
Fixtures

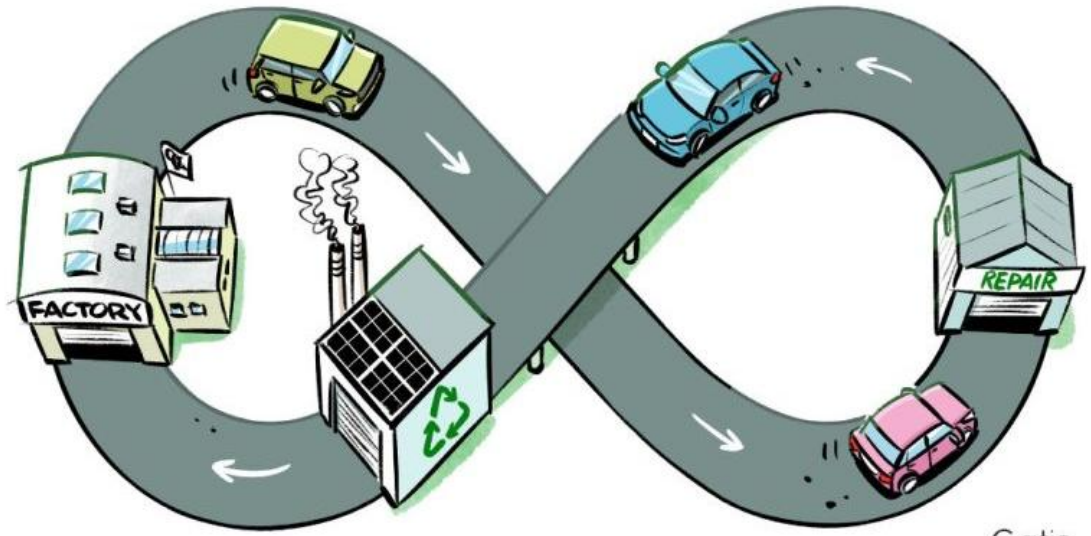
Electrical Conduit

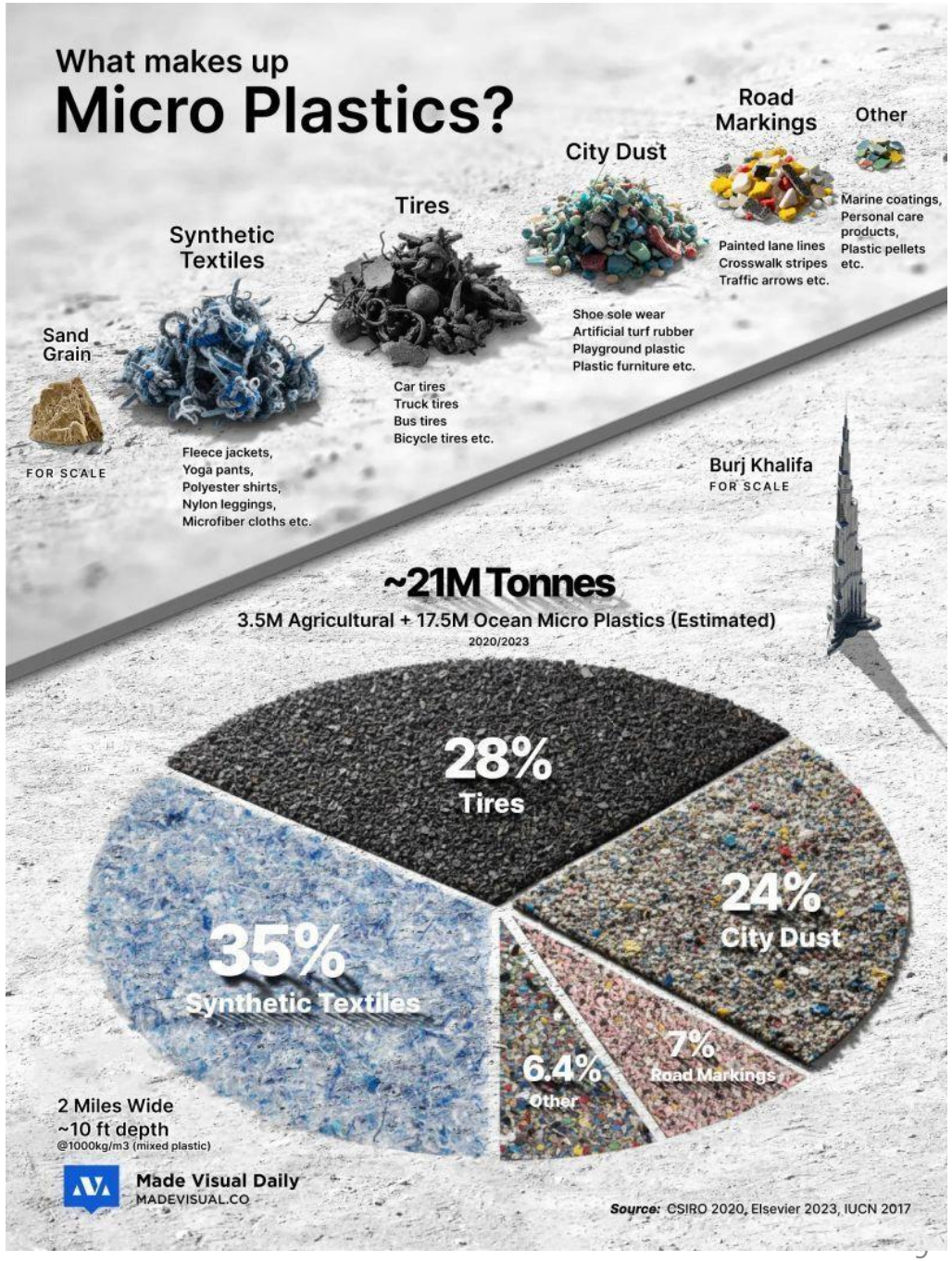

Flooring

SINGLE-USE ITEMS
PLASTIC BAGS
STIR STICKS
SIX-PACK RINGS
PLASTIC UTENSILS
SOME FOOD WARE
PLASTIC STRAWS
REUSABLE ALTERNATIVES

How Long Single Use Plastic Items Will Be Around
Plastic Bottle
450 Years
Plastic Bag
20 Years
Cigarette Filter
10 Years
Strofoam
Forever
Coffee Cup
30 Years
Plastic Straw
200 Years
Plastic Cutlery
100+ Years


FACTORY
OLD SYSTEM


FACTORY
REPAIR
NEW EU RULES
Gatis
Šlūka


What makes up
Micro Plastics?
Road Markings
Other
City Dust
Tires
Synthetic Textiles
Sand Grain
Marine coatings,
Personal care products,
Plastic pellets etc.
Painted lane lines
Crosswalk stripes
Traffic arrows etc.
Shoe sole wear
Artificial turf rubber
Playground plastic
Plastic furniture etc.
Car tires
Truck tires
Bus tires
Bicycle tires etc.
FOR SCALE
Fleece jackets,
Yoga pants,
Polyester shirts,
Nylon leggings,
Microfiber cloths etc.
Burj Khalifa
FOR SCALE
~21M Tonnes
3.5M Agricultural + 17.5M Ocean Micro Plastics (Estimated)
2020/2023
28%
Tires
24%
City Dust
35%
Synthetic Textiles
7%
Road Markings
6.4%
Other
2 Miles Wide
~10 ft depth
@1000kg/m3 (mixed plastic)
Made Visual Daily
MADEVISUAL.CO
Source: CSIRO 2020, Elsevier 2023, IUCN 2017

Degradation rates of plastics

The polymer degradation rate, r_d, is the differential mass loss per unit time

$$r_d = -\frac{dm}{dt} = k \cdot SA$$

Table 1. Estimated Half-Lives of Common Plastic Items[a]

Plastics type	Common applications	Typical thickness (μm)	Estimated Specific Surface Degradation Rate (min–max; μm year^{-1})[b]			
			Land (buried)	Land (accel. by UV/heat)[c]	Marine	Marine (accel. by UV/heat)[c]
1. PET	Single-use water bottle	500	0	–	–	110
2. HDPE	Plastic bottles	500	1.0 (0.91–1.1)	1.3 (0.55–2.6)	4.3 (0–11)	9.5 (4.5–22)
2. HDPE	Pipes	10,000	1.0 (0.91–1.1)	1.3 (0.55–2.6)	4.3 (0–11)	9.5 (4.5–22)
3. PVC	Pipes	10,000	0	–	–	–
4. LDPE	Plastic bags	100	11	22 (1.6–83)	15 (0–37)	10 (9.0–12)
5. PP	Food storage container	800	–	0.51	7.5	4.6
6. PS	Insulating packaging	20,000	0	–	–	–
7. Others	Biodegradable plastic bag	100	270 (20–1400)	320	16 (7.5–29)	180

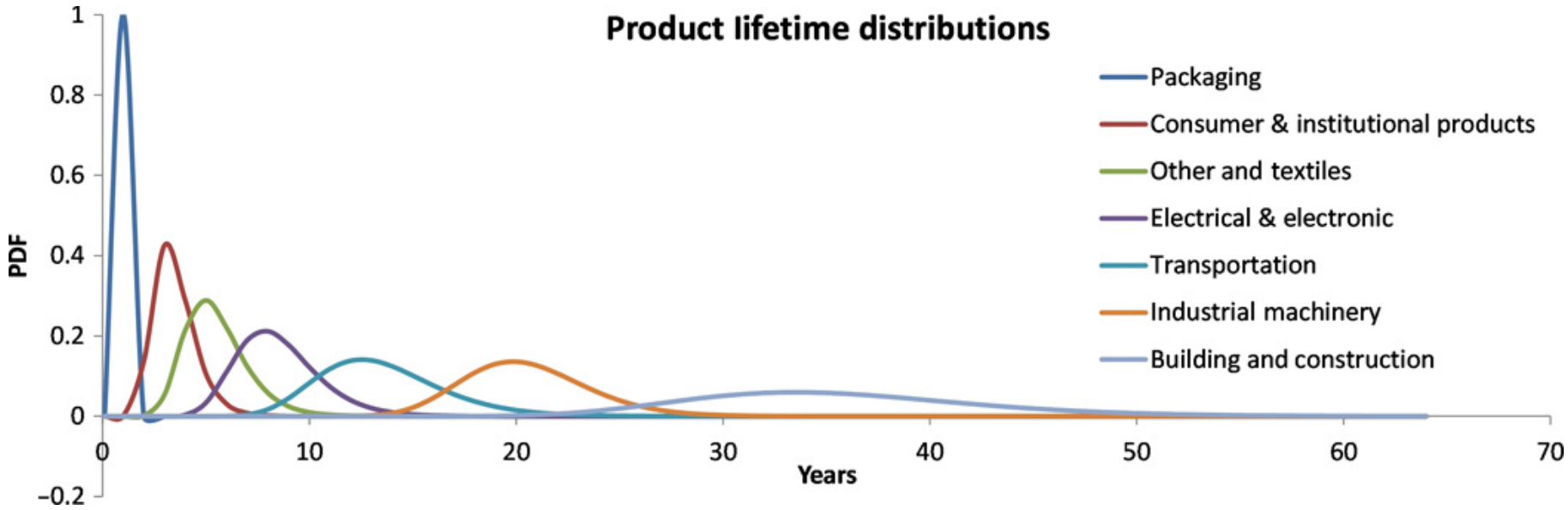
https://www.epa.gov

https://pubs.acs.org/doi/epdf/10.1021/acssuschemeng.9b06635

Organic Chemicals in the Environment

Fate of polymers after use

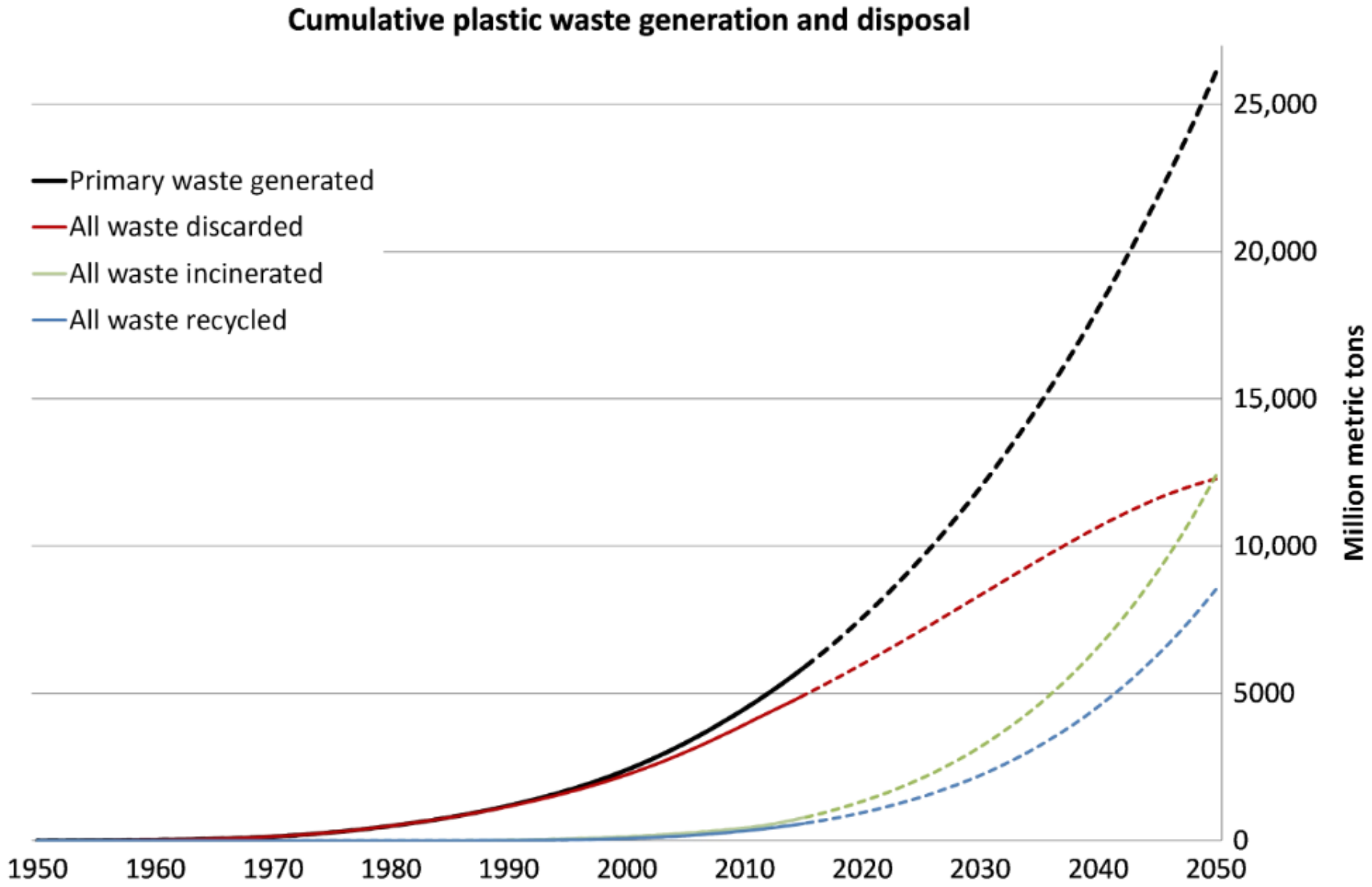
How long plastics are in use before they reach the end of their useful lifetimes



At current growth rates, the accumulation of plastics waste in landfills and/or in the natural environment is projected to reach nearly 12,000 Million tons globally by 2050

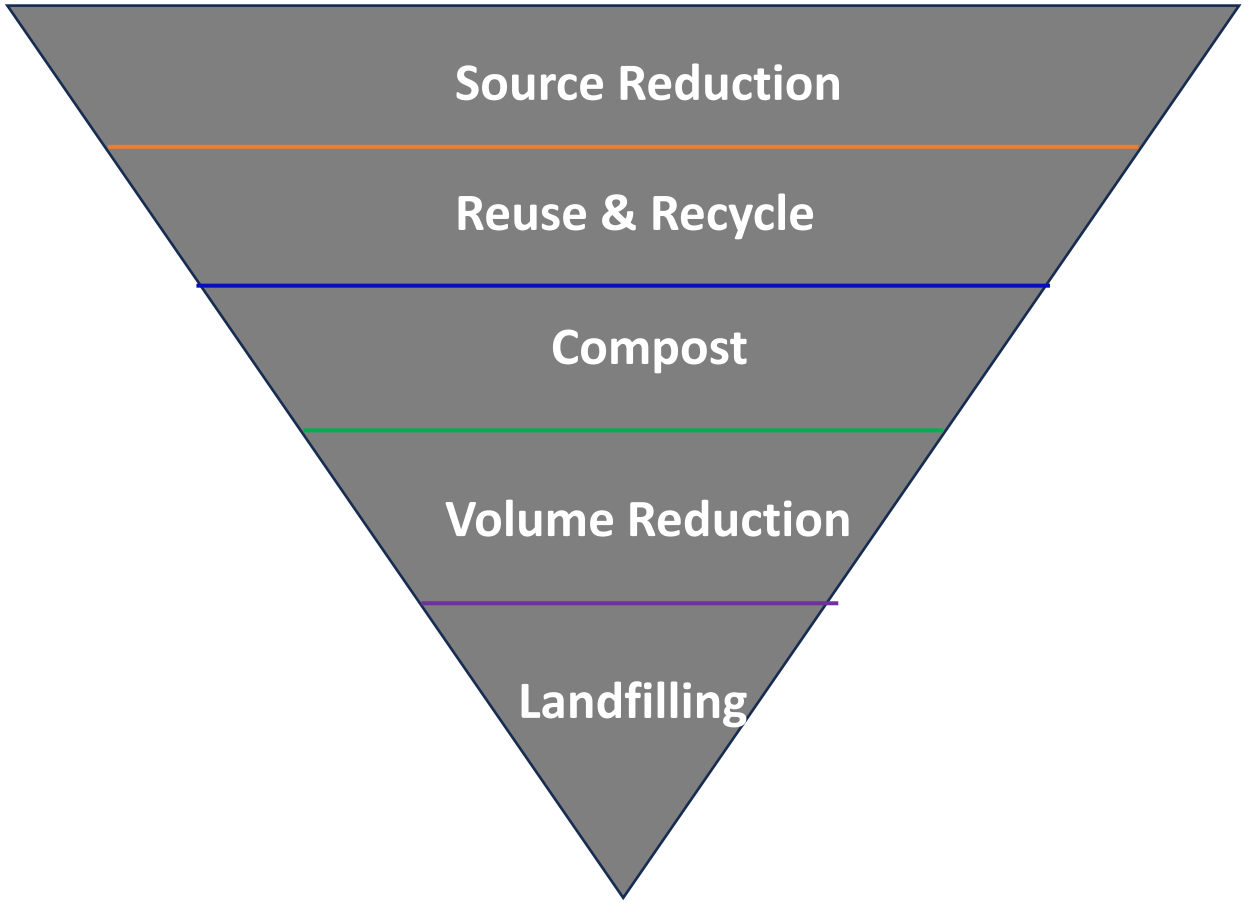
Reference: 10.1126/sciadv.1700782

Fate of polymers after use



Reference: https://dx.doi.org/10.1021/acssuschemeng.9b06635?ref=pdf

Waste Hierarchy



Source Reduction

- By reducing or eliminating the generation of harmful and persistent wastes, waste minimization supports efforts to promote a more sustainable society.
- Waste minimization involves redesigning products and processes and/or changing societal patterns of consumption and production

How to do it?

- The best way to deal with trash is to not have any!
- Reducing the amount of trash you have to throw out actually prevents waste from piling up in the first place.
- To reduce your waste, avoid unnecessary packaging and items designed to be used only once.
  - Reduce the need for 'single use' plastic bags by bringing your own bags when you shop and use a travel mug when you buy coffee.
  - Choose durable, reusable products to make less trash.

Reuse & Recycle

- Reusing items can sa
- Extend the life of iten
  - For example, reu
    goods and clothi

TOI City Hyderabad Mumbai Delhi Bengaluru Kolkata

TRENDING Meerut Murder Yashwant Verma Judge Sushant Singh Rajput case Train Acci

THIS STORY IS FROM NOVEMBER 21, 2024

IIT-Hyderabad develops tech to transform plastic waste into products

U Sudhakar Reddy / TNN / Nov 21, 2024, 19:19 IST SHARE AA FOLLOW US

Researchers at IIT Hyderabad have found a way to transform plastic bottles into useful materials. They used special catalysts made from manganese oxide and niobium pentoxide to break down polyethylene terephthalate (PET) into chemicals that can be used to make biodegradable plastics and resins. This breakthrough could help reduce plastic waste and our dependence on fossil fuels.

Read Less

HYDERABAD: In a potential solution to the problem of plastic waste, researchers at the Indian Institute of Technology Hyderabad (IITH) have developed a sustainable technology

Recycle

- Every day we use products made from recycled materials.
- Take your glass, cans, newspapers, milk jugs and other acceptable recyclable items to your local transfer station, drop off location or place out for curbside collection so that they can be turned into new products like jackets
- Frisbees, paper products, and soda cans. Recycling saves money, energy, and the environment.
- 48% of municipal solid waste was recycled in 2015.

Mobiles sets

Old cars

Compost

- Composting is nature's way of recycling organics.
- When you compost, you convert vegetable scraps, leaves, grass clippings and other materials into a nutrient rich soil material.
- You can use finished compost in your garden and around shrubs or other plants to help them grow.
- Composting also reduces the amount of materials that need to be disposed of, reducing those related costs.

Processing and Beneficial Use

- Processing reduces the volume of materials to be landfilled and can create products such as fuel oils and steam for electricity generation.
- Beneficial use means the reuse of solid waste as a substitute for raw material in manufacturing, as construction material or fill, as a fuel, or as an agronomic soil amendment.

Solution to plastic pollution

Prof. R. Vasudevan

Plastic has slowly became an integral part of all human requirements. Plastic carry bags, packaging material, bottles, cups, and various other items have slowly replaced everything made of other material due to the advantages of plastic.

Plastic is durable, easy to produce, lightweight, unbreakable, odorless, and chemical resistant.

Disadvantages

- Leaching of microplastics into soil and water bodies
- Accumulation by aquatic life

http://timesofindia.indiatimes.com/articleshow/84883086.cms?utm_source=contentofinterest&utm_medium=text&utm_campaign=cppst

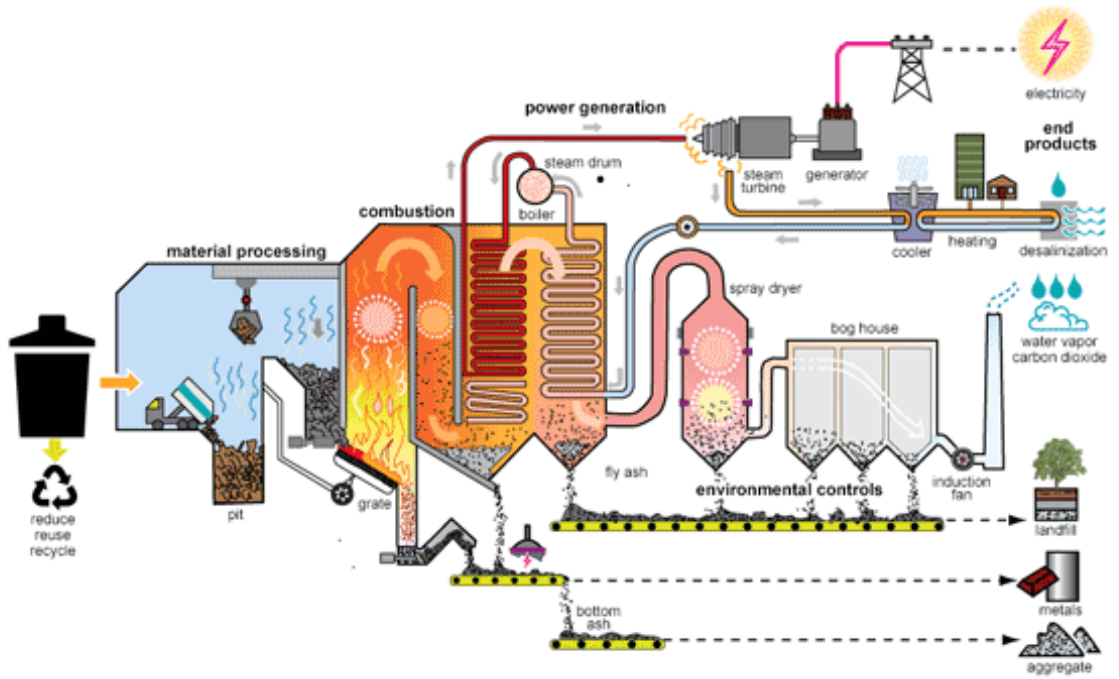
Waste-to-Energy

Waste-to-Energy facilities accept our solid waste and combust it at very high temperatures, producing heat that is used to convert water into steam. The steam is used to run turbines that generate electricity.

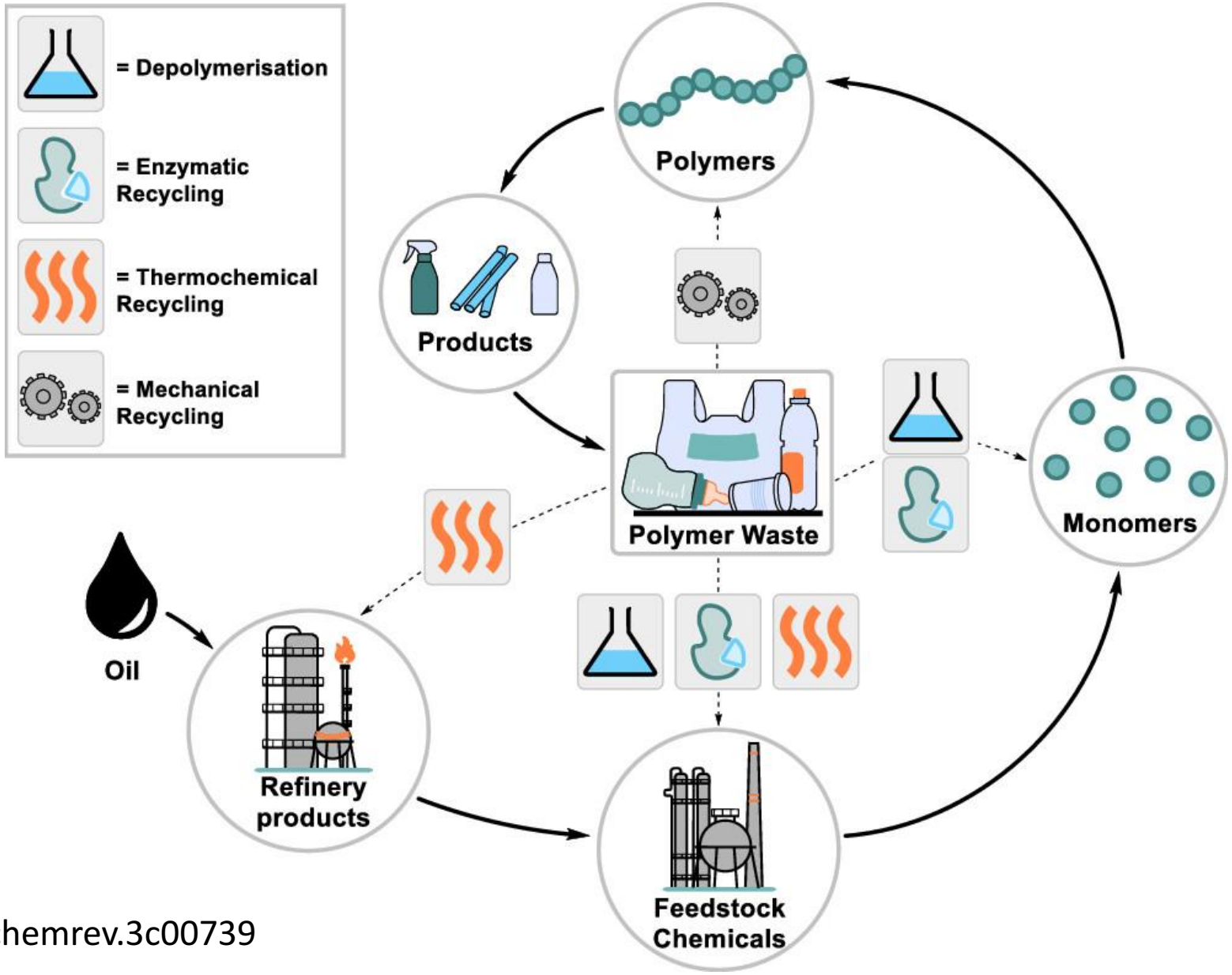
Vienna incineration plant

7 stages

1.Waste is dumped into a large pit.
2. dumps it in a combustion chamber.
3.The waste (fuel) is burned, releasing heat.
4.The heat turns water into steam in a boiler.
5.The high-pressure steam turns the blades of a turbine generator to produce electricity.
6.An air-pollution control system removes pollutants from the combustion gas before it is released through a smokestack.
7.Ash is collected from the boiler and the air pollution control system.



Waste-to-Wealth Depolymerization



https://doi.org/10.1021/acs.chemrev.3c00739

Fuel from plastics

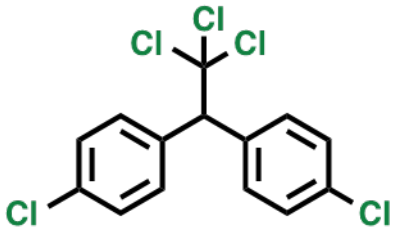
- Pyrolysis
- Thermal decomposition
- Catalytic converters (Zeolites and cobalt)

Landfill

- Today's landfills are very different from the old ones where people just dumped their garbage in an open area.
- Landfills are constructed and operated to strict environmental standards, including liners to protect groundwater.
- Within this hierarchy, landfilling waste is the lowest priority of the solid waste management options.

Waste Minimization: Process Manipulation

Many industries are also purifying wastes to recover chemicals that can be reused, recycled, or sold to another industry for use as raw materials.



DDT (Dichlorodiphenyltrichloroethane)

HCl

In Europe, several countries have now established waste exchanges, where waste chemicals are successfully traded

Organic Chemicals in the Environment

Fate of polymers after use

- It has been estimated that plastics kill or injure tens of thousands of sea birds, seals, sea lions, and sea otters each year, and hundreds of whales, porpoises, bottlenose dolphins, and sea turtles

- A large portion of the reported fatalities are due to the entanglement of birds and mammals in the drift nets of commercial fishing boats

Possible solutions

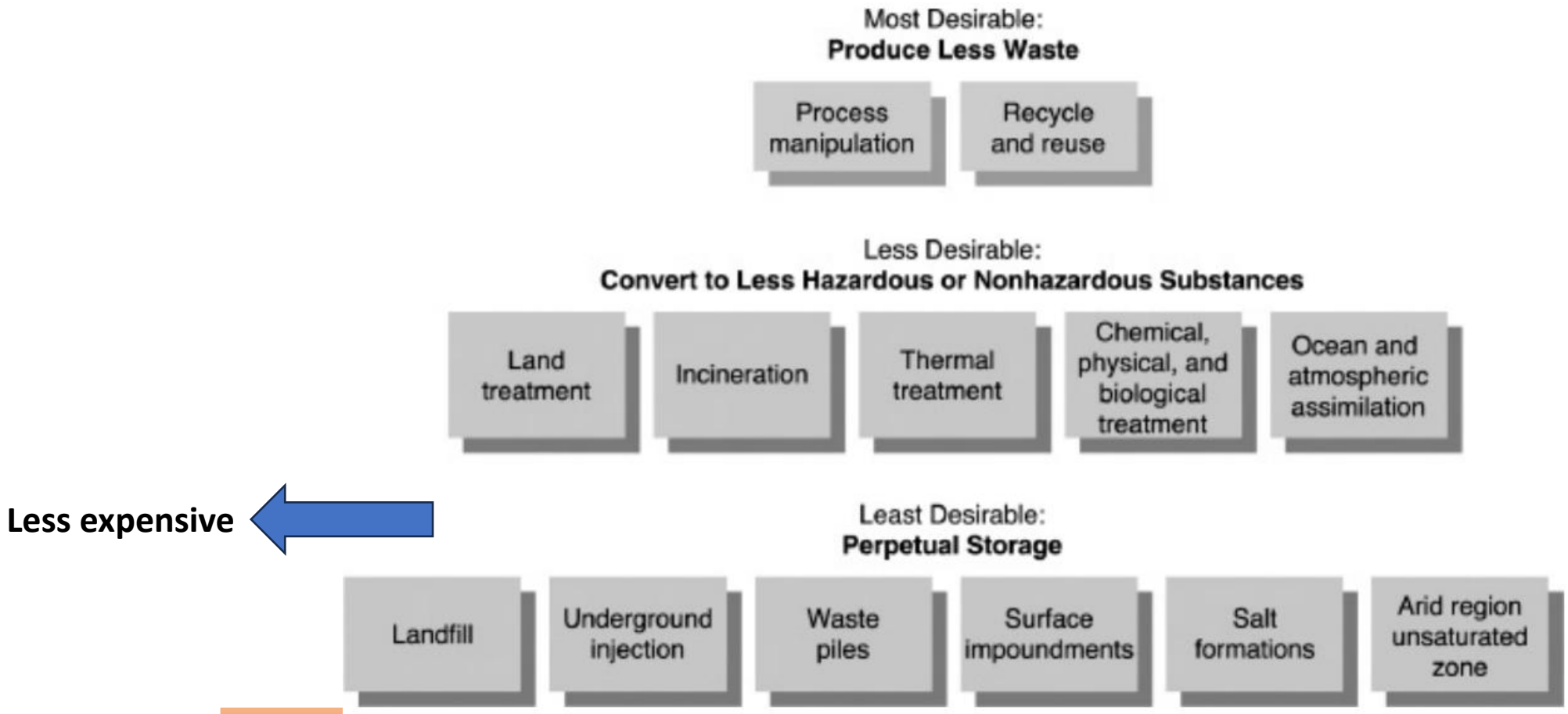
- Development of biodegradable polymers
- Increase recyclability
- Reducing the usage of single use plastic

Protect of our Planet Earth

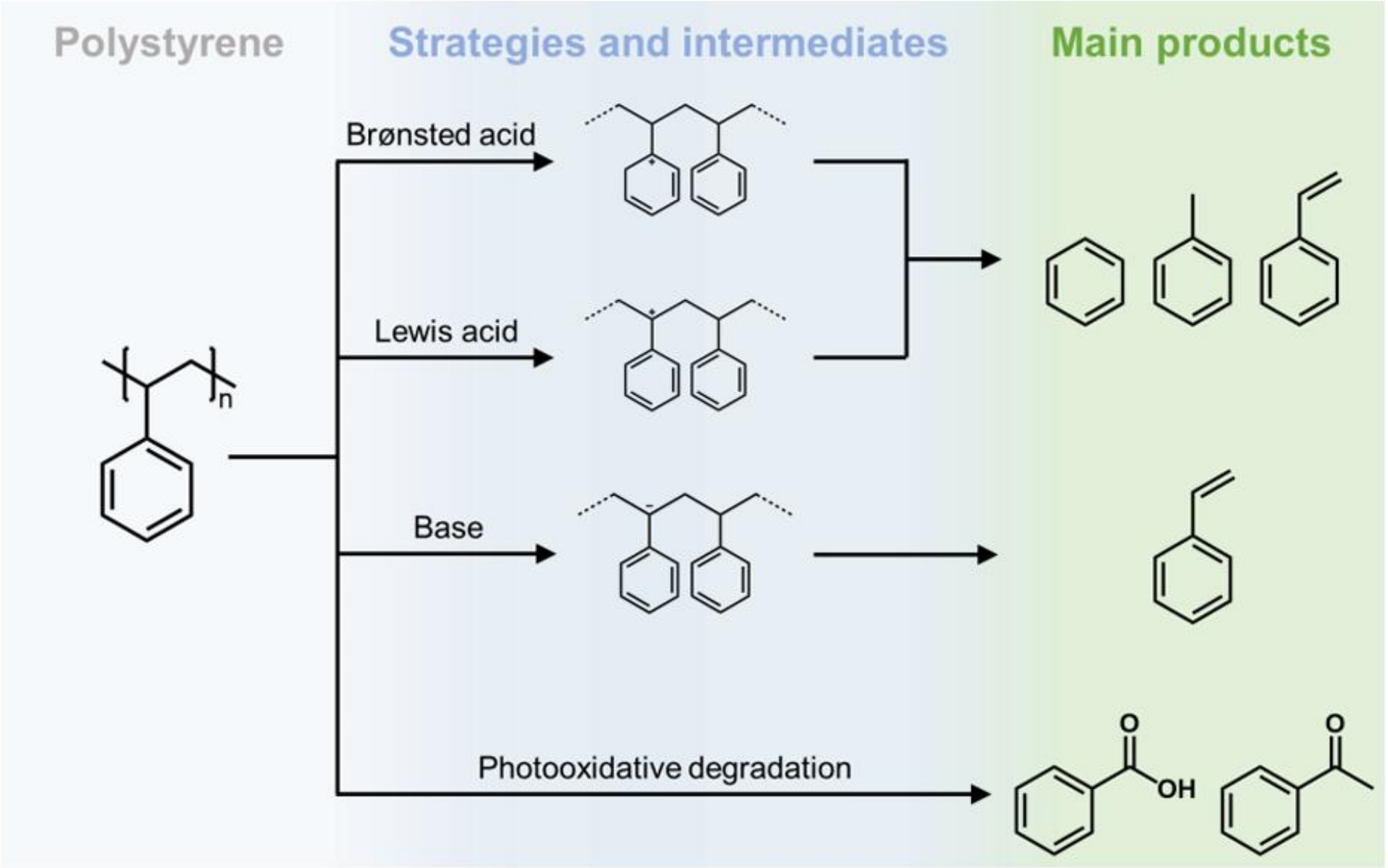
- ✓ Conserve water
- ✓ Conserve Energy
- ✓ Grow more trees
- ✓ ‘RRR’ – waste
- ✓ Ban Polyethylene
- ✓ Use Renewables

Policy for Management and Disposal of Hazardous Waste

To minimize the amount produced,
(2) to convert the waste to a less hazardous or non hazardous form, and
(3) to isolate the waste in a secure, perpetual-storage site

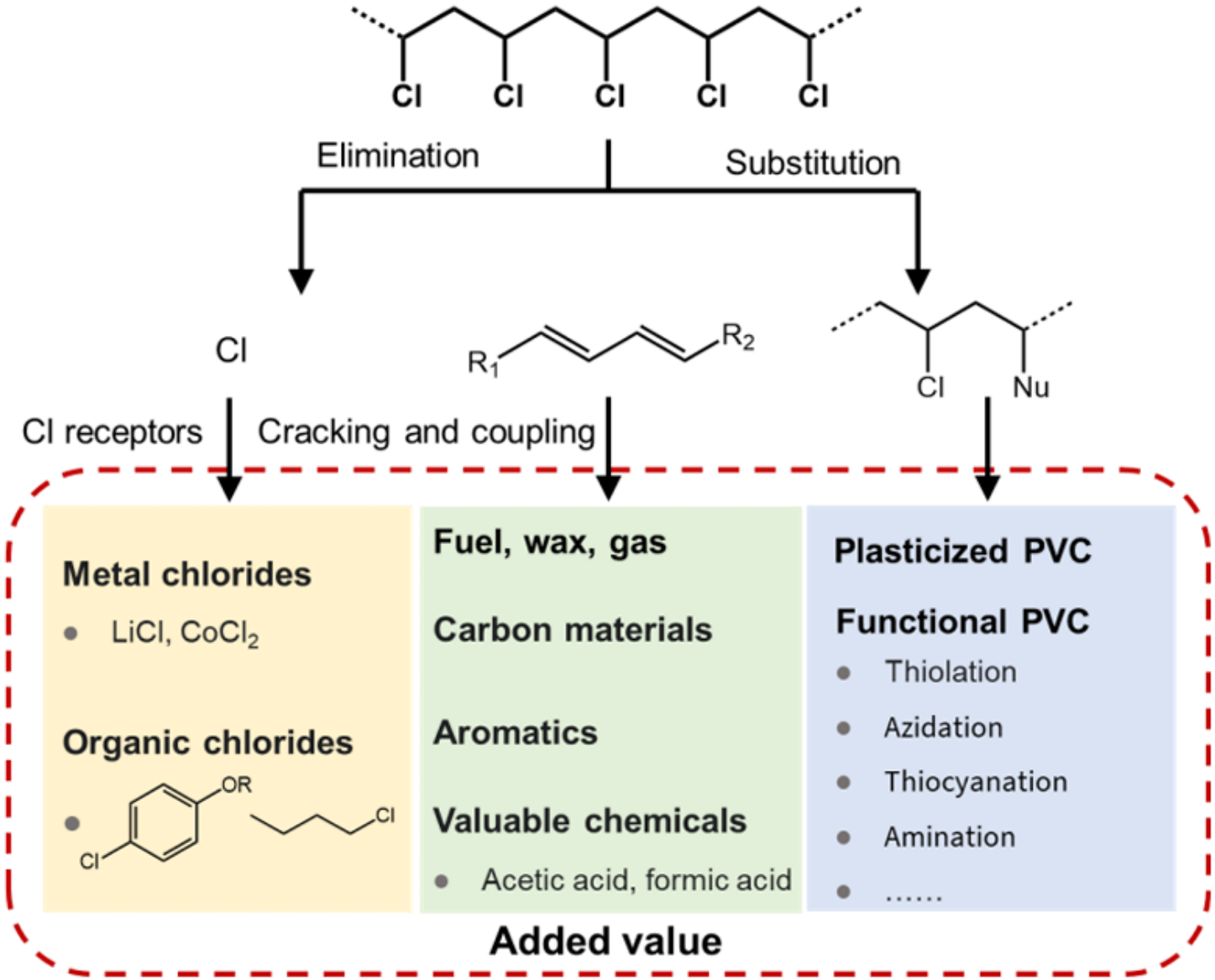


Chemical upcycling strategies of polystyrene and its main products



Energy and Environment **2025**, *2* (1), 4. https://doi.org/10.53941/see.2025.100004

Value-added upcycling strategies for PVC



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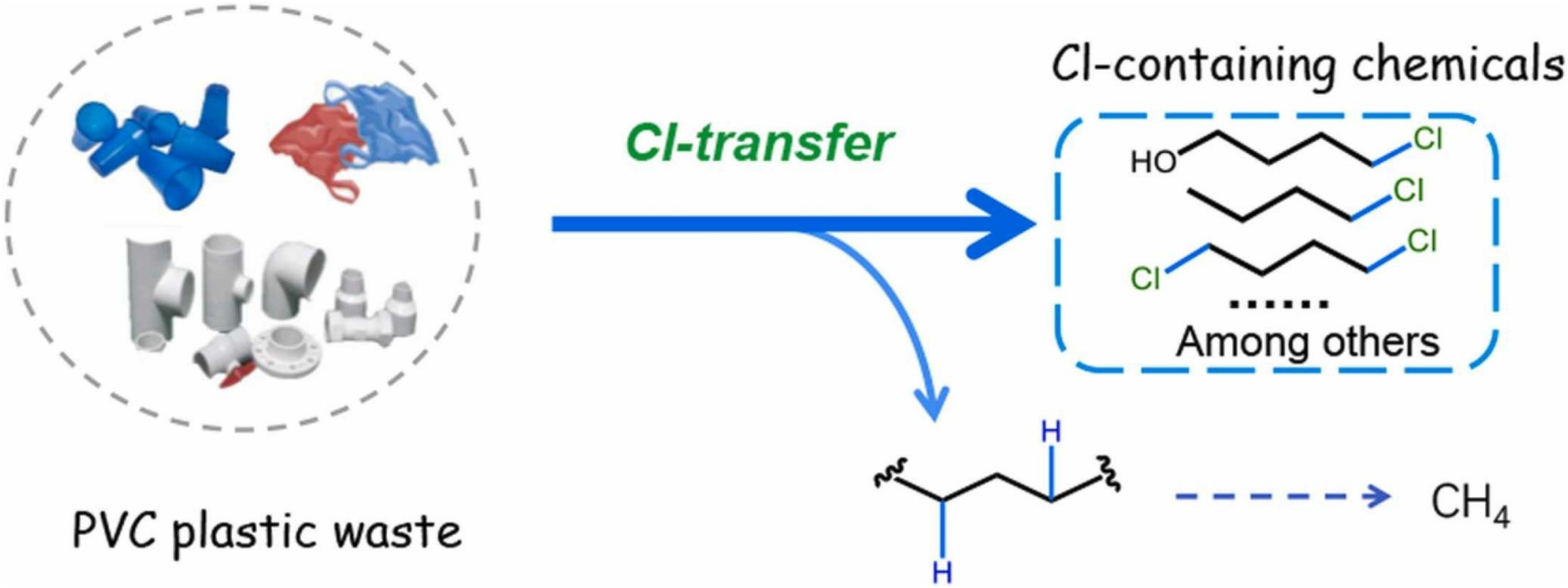
Value-added upcycling strategies for PVC



https://doi.org/10.1016/j.apcatb.2023.122671